GESPYE GESTIÓN PÚBLICA Y EMPRESARIAL (e-ISSN: 3061-8630)
Año 3, Núm. 6 (jul-dic 2026) • Págs. 19-39 • DOI https://doi.org/10.32870/gespye.vi6.2
Recibido: 16/01/2026 • Aprobado: 15/05/2026 • Publicado: 20/06/2026

Artículos

Citizen participation and the use of AI to improve governance guidelines

Participación ciudadana y el uso de la IA para mejorar las directrices de gobernanza

Cinthya Karina Camacho-Sotelo[2]
Adriana Vázquez Yáñez[3]
César Lugo Valdez[4]

Abstract. The objective of this research is to describe the digital evolution of governments through the implementation of various technologies, such as artificial intelligence (AI), and to examine how this has contributed to enhancing citizen participation. Governments face two primary challenges: the first is to improve trust with citizens and encourage their participation in government decisions, and the second is to utilize technologies that facilitate this interaction. Currently, countries have made progress on AI issues, primarily in regulating it. However, the use of AI in citizen participation is still in its early stages, as it is mainly employed for citizen consultations regarding government information, procedures, and government outreach to citizens on specific issues. Still, it has not advanced to the point where governments encourage citizens to be more involved in collaborating on all stages of a consultation, even after it is implemented.

Keywords: Artificial intelligence, Citizen participation, Governance

Resumen. El objetivo de esta investigación es describir la evolución digital de los gobiernos mediante la implementación de diversas tecnologías, como la inteligencia artificial (IA), y examinar cómo esto ha contribuido a mejorar la participación ciudadana. Los gobiernos se enfrentan a dos desafíos principales: el primero es mejorar la confianza con los ciudadanos y fomentar su participación en las decisiones gubernamentales, y el segundo es utilizar tecnologías que faciliten esta interacción. Actualmente, los países han avanzado en materia de IA, principalmente en su regulación. Sin embargo, el uso de la IA en la participación ciudadana aún se encuentra en sus primeras etapas, ya que se emplea principalmente para consultas ciudadanas sobre información, procedimientos y comunicación gubernamental con los ciudadanos sobre temas específicos. Todavía no ha avanzado lo suficiente como para que los gobiernos fomenten una mayor participación ciudadana en todas las etapas de una consulta, incluso después de su implementación.

Palabras clave: Inteligencia artificial, Participación ciudadana, Gobernanza

Introduction

Governments today depend, more than ever, on the participation of their citizens, so that their voices and opinions can help shape public policies that are better aligned with real

[2] University of Guadalajara, email: cinthya.camacho@cucea.udg.mx, Orcid: https://orcid.org/0009-0008-3369-4287
[3] University of Guadalajara, email: adriana.vazquez0217@alumnos.udg.mx Orcid: https://orcid.org/0009-0002-3151-391X
[4] University of Guadalajara, email: cesar.lugo2225@alumnos.udg.mx Orcid: https://orcid.org/0009-0006-9878-6989

needs. Through citizen participation, governments seek to analyze those needs from the perspective of the people who experience them daily, supporting more informed decision-making. This participation has limitations, however, as it has been used primarily as a consultative measure. Consulting citizens—despite its value—poses the challenge of evaluating the contributions received, with implications for the use of resources, including human resources (Romberg & Escher, 2024).

Criado (2016) notes that public management is undergoing transformations similar to those occurring in other political, social, and economic dimensions, and that this change is closely tied to technological advances and their repercussions across society. One of the tools driving the evolution toward digital governments is artificial intelligence (AI), increasingly used and ever more present in public discourse (Palak Dev & Sainger, 2025). In the public sector it is used to provide better services (Sharma & Padashetty, 2025) and to transform internal processes (Mellouli et al., 2024).

Governments have historically sought greater efficiency, including in citizen participation, and have undergone several digitalization-focused transformations (Puerta et al., 2023). After electronic government and e-government 2.0—which incorporated social networks into public operations—we are entering e-government 3.0, where tools such as artificial intelligence, blockchain, and virtual and augmented reality are already transforming the delivery of public services and governance (Vrabie, 2023). Within this process of public innovation, digitalization and digital transformation generate value for governments and citizens, with digital government and open data as central axes (Ospina & Zambrano, 2023).

Artificial intelligence is changing how governments interact with citizens, manage information for decision-making, and address social challenges in what is termed interactive governance, allowing citizens to be involved more effectively in policy formulation and service delivery. AI can process the large volumes of information generated through participation, improving both the quality of decisions and citizen engagement (Pislaru et al., 2024, citing Zuiderwijk et al., 2021). As Pislaru et al. (2024) observe, AI's capacity to optimize administrative processes reduces inefficiencies and allows public officials to focus on strategic, citizen-centered decisions.

The purpose of this research is to analyze how citizen participation, supported by artificial intelligence, has been implemented in various governments that have adopted it. Being a relatively new and evolving phenomenon, it requires constant analysis. We therefore seek to identify the main success factors of certain countries in applying AI in their governments, as well as the limitations of AI implementation in citizen participation in the context of specific countries and cities.

What is citizen participation?

Participation is a flexible term whose definition varies according to context, and it can be categorized by the sense in which it is interpreted—sharing or exchanging information, inclusion in decision-making, and the exercise of influence and control. From the perspective of sharing or exchanging, it implies the active involvement of stakeholders in discussing ideas, dedicating time and resources, making decisions, and taking joint action to achieve a development objective (Mohammed, 2013). From a citizenship approach, participation is a form of self-expression: beyond being contributors or customers, citizens act as critics and shapers of the policies that affect their lives, a role made more relevant by the growth of the third sector and increasing public–private collaboration (Lahdili & Nyadera, 2024).

Citizen participation can also be defined as the actions and initiatives of citizens and organizations aimed at influencing policies and governance at various levels (Citizen Participation Forum, 2021). From a practical standpoint, Nabatchi and Amsler (2014) define it as the use of face-to-face and online methods to bring people together to address significant issues. The OECD (2017) points out that participation can involve several actors, understanding citizenship broadly as all inhabitants of a given place—regardless of age, gender, religion, or political affiliation—as well as other interested or affected parties, whether governmental institutions, civil society, academia, media, or the private sector.

Several agencies distinguish participation by levels. The OECD (2022) identifies three: (a) information, a unilateral relationship where government provides data to citizens; (b) consultation, a bilateral interaction where citizens and stakeholders provide feedback on specific problems; and (c) active participation, which involves collaboration throughout the policy cycle, allowing citizens to influence the agenda and proposals, although the final decision remains with government. Similarly, the International Association for Public Participation (IAP2, 2024) defines five levels: inform, consult, involve, collaborate, and empower—the last delegating the final decision to citizens.

Jäntti and Kurkela (2021) distinguish two forms of participation: institutional, where municipalities act as spaces that facilitate participation, and non-institutional, such as protests or autonomous civil-society actions, where municipal influence is limited. Jäntti et al. (2023) identify three trends explaining how participation has been integrated into public-sector organizations: as part of government openness to strengthen transparency and democracy; as a reaction to the crisis of representative democracy; and as a strategy to address complex social problems through collaboration. They add three levels of governance for embedding participation: the strategic level (vision and policy objectives), the executive level (management practices requiring human, financial, technological, and knowledge resources), and the citizen-interface level (where participatory instruments connect citizens with government).

Citizen participation can take many forms—councils, forums, surveys, collaborative workshops, and participatory budgets—which are part of a more complex network of processes that also influence local government decisions (Nabatchi & Amsler, 2014; Jäntti & Kurkela, 2021). The dialogue between citizens and the state is sustained by democracy, so it is useful to recall the classifications found in democratic theory: representative, participatory, deliberative, and radical. In practice, no democracy is confined exclusively to one model (Contreras & Montecinos, 2019).

Giffinger et al. (2007) consider citizen participation an element of good governance, increasingly central to debates on democracy and development. Bingöl (2022) argues that participation enables citizens to influence how information is disseminated, how goals and policies are defined, and how economic resources are allocated, while Lukensmeyer (2013) describes a process through which citizens learn, express opinions, find common ground, and influence state decision-making. Common approaches include the electoral, the exchange of legislative and administrative information (such as public hearings), civil-society participation through volunteering and associations, and deliberative approaches based on consensus (Lindgren et al., 2019). Collaborative models require technological infrastructure—platforms, applications, and tools—not only to share information but to improve participation in debates on critical public issues, policy implementation, and service delivery (Zheng et al., 2014; Giffinger et al., 2007). Technology facilitates communication across levels while reducing the costs of participation, especially time and distance (Castells, 2011); yet, if participation processes are not carefully analyzed, they can easily become a technocracy—a government of experts in which participation is superficial and one-way (Lan et al., 2021; Fischer, 1990).

The evolution of citizen participation toward the digital

Realist democratic theory underpins participatory democracy, which emphasizes citizen participation in decision-making at all levels of government, complementing both representative institutions and direct-democracy mechanisms such as referendums (Balduzzi & Siclari, 2024). Democratic theorists have proposed democratic innovations—processes or institutions designed to enhance the role of citizens in governance and expand opportunities for participation, deliberation, and influence (Elstub & Escobar, 2019). To foster a positive perception of smart policies, governments must integrate smart technologies into their communication and participation strategies through accessible, clear, and high-quality media (Hartley, 2023).

The use of digital tools to foster participation—known as e-participation or digital participation—has expanded globally (Steinbach et al., 2019). Many large cities already drive participation through information and communication technologies (ICTs) and new media such as social networks and virtual platforms (Bonsón et al., 2015; Gilman & Peixoto, 2019). Digital tools can facilitate citizen interaction and self-organization, reduce the costs of consulting the population, lower barriers to participation, promote inclusion and equity, and enable direct links between citizens, politicians, and policymakers (Hovik et al., 2022; Ruano & Reichborn-Kjennerud, 2022). Technologies such as AI, blockchain, and virtual reality can reduce barriers, increase government capabilities, and empower citizens through more intelligible and accountable participatory processes (OECD, 2025a).

Digital transformation seeks to integrate information technologies to enhance government services—including communication and participation—while considering factors such as residential environment, employment, income, and education that influence the impact of AI on citizen–government communication (Pislaru et al., 2024). The first global initiatives implemented simple online services: providing information to promote transparency, enabling fee payments, and offering access to public information through web portals (Brown, 2005). E-government continues to evolve with technological advances, making services more efficient through AI, blockchain, and other innovations (Lykidis et al., 2021).

Within public administration, the limits of approaches such as New Public Management have prompted a turn toward collaborative governance, in which values such as innovation, transparency, and participation are promoted, with citizen participation as a key component (Mariñez, 2022). Participation is thus increasingly relevant to policymaking, driven by information technologies and AI, which facilitates dialogue with administrators, optimizes decision-making, and processes large volumes of data to identify problems and forecast policy impacts (Lahdili & Nyadera, 2024). AI can revolutionize public management, enabling authorities to respond better to citizens' needs and strengthening trust in public services (Vrabie, 2023); by fostering collaboration and responsiveness, it enhances engagement through platforms that allow citizens to express concerns, provide feedback, and contribute to policy formulation (Pislaru et al., 2024).

AI's relationship with democracy is complex: it can threaten democratic legitimacy (Danaher, 2016) and erode access to truthful knowledge (Coeckelbergh, 2022), but it can also improve the policy cycle and facilitate mass deliberation (Landemore, 2022). States must address difficulties such as economic crises, poverty, and emerging diseases; during the COVID-19 pandemic, many governments developed applications to monitor the spread of the virus and inform containment strategies (Ospina & Zambrano, 2023). In recent years, countries have advanced on digital government and AI, monitored internationally through indices on digital government, open data, and AI readiness, generating public value through more efficient services aimed at improving citizens' quality of life (Ospina & Zambrano, 2023).

Citizen participation and the use of artificial intelligence

The increasing adoption of AI in government is creating numerous opportunities for public administrations, with the potential to accelerate the transformation of service delivery and policy design within public-sector ecosystems (Zuiderwijk et al., 2021). Its incorporation has been welcomed as a way to strengthen governments' operational capacity; AI now supports decision-making, predictions, and the handling of complex challenges. In low- and middle-income countries, these technologies help optimize service delivery, facilitate interaction with citizens, strengthen democracy, and overcome language barriers that limit the exercise of participation rights (Zidouemba, 2025).

The use of AI-driven tools in decision-making, policy implementation, and government interaction—known as algorithmic governance—is already integral to the contemporary administrative state (Engstrom et al., 2020). Algorithms, the centerpiece of AI, allow public administrations to access more efficient, economical, or apparently neutral solutions by analyzing large volumes of data (Li et al., 2023). Applications extend to sectors such as transportation (Kouziokas, 2017), mental health services (Zhu et al., 2022), the criminal justice system (Whitford et al., 2020), and policing (Meijer et al., 2021).

Zuiderwijk et al. (2021) group the benefits of applying AI in government into nine areas: improvements in efficiency and performance; capabilities to identify and monitor risks; economic advantages; optimization of data and information processing; improvements in service delivery; benefits for society; support for decision-making; strengthening of citizen engagement; and contributions to sustainability. Adoption varies by context, ranging from data analytics for complex decisions to fraud detection, risk management, mobility planning, health services, environmental protection, and the automation of administrative functions (Mellouli et al., 2024). AI is also applied to public-health challenges, such as controlling Zika and dengue, to help governments monitor their spread and act in a timely manner (Zidouemba, 2025).

Studies by Engstrom et al. (2020), Tangi et al. (2022), and van Noordt and Misuraca (2022) classify government uses of AI in areas such as policy formulation, service provision, internal human-resource management, and citizen engagement. The European Commission's Adopt AI Study (2024) found AI prominently implemented in general public services, economic affairs such as transportation, security and public order, and health. AI can improve processes and procedures, help meet strategic objectives, reduce costs and environmental impacts, combat fraud and abuse by enhancing oversight of public funds, increase efficiency, and support informed decision-making (Fernández, 2023).

Artificial intelligence has transformed how governments serve citizens, requiring changes in the design and delivery of public services (OECD, 2025b). The COVID-19 pandemic marked a watershed, compelling governments to keep operating under difficult circumstances and improving the responsiveness and reliability of the public sector (OECD, 2023b). For these changes to be sustainable, governments must improve their digital public infrastructure and focus on emerging technologies—AI, automation, and modular code—to expand their innovative and digital capabilities (OECD, 2023b; OECD, 2024). One application of this digital transformation is citizen participation, where AI plays a crucial role in promoting effective, transparent governance and helping citizens propose innovative ideas to solve complex problems.

Challenges for governments in the use of AI

Despite its benefits for informed decision-making, irresponsible use of AI can lead to discriminatory practices and violations of data-protection regulations. Insufficient digital literacy, cyber-attack risks, threats to privacy and civil liberties, high implementation costs, and shortages of trained personnel are significant risks that may hinder the effective use of AI in participation mechanisms (Lahdili & Nyadera, 2024). Data protection and privacy raise concerns about how AI systems collect, store, and manage personal information, so transparency and responsible data use are critical to strengthening public trust (Gesk & Leyer, 2022). Citizens also fear that algorithms may reinforce existing inequalities if adequate regulation and oversight are lacking, making transparency and fairness essential for public acceptance (Madan & Ashok, 2022; Chen et al., 2021).

Zuiderwijk et al. (2021) group the challenges of applying AI in government into eight categories: data-related problems; organizational and managerial difficulties; skills and technical-capacity gaps; difficulties in interpreting AI-generated results; ethical and legitimacy issues; political, legal, and regulatory obstacles; social and community impacts; and economic implications. Lahdili and Nyadera (2024) warn that over-reliance on AI poses risks to political, social, economic, and cultural domains, including the inability to replace human attributes such as creativity and empathy, the danger of elites using AI to control or repress society, and the commercialization of personal data. In low- and middle-income countries, where digital literacy is low and skilled personnel scarce, AI could worsen job losses (Zidouemba, 2025). Engstrom et al. (2020) identify six implications of public-sector AI: difficulties in developing institutional capabilities; challenges in ensuring transparency and accountability; unintended biases; risks to citizens' procedural rights; threats from adversarial learning; and the need to complement technical expertise through specialized recruitment.

Mellouli et al. (2024) classify these challenges into two levels. At the macro level (political and economic), threats include unemployment from job replacement, high energy consumption, the spread of misinformation, high implementation costs, and the loss of human knowledge. At the micro level (policy formulation and decision-making), risks include biases in training data, replication of human error, systemic discrimination, the inability to detect bias or exclusion, unfair decisions, reduced inclusiveness, privacy violations, lack of transparency, and declining public trust. Effective governance of data and technologies is therefore essential to mitigate these risks.

Through research with European Union countries, the OECD (2025a) identified the most urgent participation challenges: a lack of trust, as citizens perceive that their participation has no impact and doubt that they have the skills to make informed decisions; the difficulty of evaluating participation, given low accountability after a process, unclear feedback, and difficulty interpreting contributions; institutional problems such as limited awareness within public administration and a lack of incentives; the exclusion of marginalized groups, since methods are often inaccessible and use overly technical language; and a gap between process and results that leaves the public feeling outside the process. The use of AI in democratic processes also poses risks—privacy issues, potential surveillance, data distortion, and a deepening digital divide that relegates illiterate and techno-skeptical groups—so governments must regulate AI and invest in secure tools that genuinely enhance participation and deliberation.

Global context of artificial intelligence in governments

Multiple reports assess governments' progress toward becoming more digital and innovative. Oxford Insights' Government AI Readiness Index 2024 ranks, among 188 countries, the ten best prepared for government AI: the United States, Singapore, Korea, France, the United Kingdom, Canada, the Netherlands, Germany, Finland, and Australia. Both the World Economic Forum (2024) and the World Bank (2022) agree that the regions most advanced in AI policy and governance are the United States, China, and the European Union, followed by Latin America, and lastly Africa, East Asia, and the Pacific.

By region, the 2024 index identifies North America (the United States and Canada) as the strongest, owing to the consolidation of the United States in the technology sector and its global leadership in AI. Western Europe ranks second, dominated by France—the region's highest scorer—followed by the United Kingdom, the Netherlands, Germany, and Finland, and standing out for its data infrastructure and consolidated governance frameworks. East Asia ranks third, with Singapore and the Republic of Korea among the top performers; China, despite its prominence in some areas, ranks lower owing to limited availability and accessibility of public information (Oxford Insights, 2024).

According to the OECD (2025b), in recent years the countries investing the most venture capital in AI have been the United States—the clear leader since 2018, peaking in 2021—followed by China, with the European Union, the United Kingdom, Korea, Germany, India, Israel, and Canada participating to a lesser extent. Measured by the share of scientific publications on AI, the leaders are China (22% of global publications), the European Union (14%), and the United States (11%), reflecting sustained investment in research and development. Not all countries, however, have the financial, political, and technical resources to develop effective AI governance (WEF, 2024): of 193 countries, 114—most in the global south—lack national AI strategies, which can give rise to privacy violations and improper uses of AI.

The World Bank's GovTech Maturity Index 2022 shows that, while countries improved online public services, digital citizen participation scored low, indicating a focus on government systems and service-delivery platforms at the expense of digital citizen feedback; the report recommends developing multifunctional platforms that deepen the citizen–government relationship and build public trust (World Bank, 2022). One area that can be strengthened through this connection is electoral participation, which has declined in many countries due to falling confidence and growing disaffection with democratic processes. In an OECD survey on perceptions of opportunity to influence local decisions, the countries scoring above the OECD average (0.41) were the Netherlands (0.53), Ireland (0.52), Canada (0.51), Mexico (0.51), the United Kingdom (0.50), New Zealand (0.48), Estonia (0.46), Denmark (0.45), Luxembourg (0.45), and Austria (0.44), confirming that active participation is more effective at the local level (OECD, 2021b).

Even so, citizens do not generally see themselves as active participants in policymaking: only 38% expect the government to act on a deficient service or implement innovative ideas, and only 30% believe their opinions on government actions are taken into account (OECD, 2021b). Technology can serve as a medium to strengthen participation and accountability: in 2021, online deliberation was the medium most used to evaluate and design public policies, and online consultations were active in 85% of OECD member countries. Civic technology helps citizens feel considered and able to express their needs, and governments should leverage emerging technologies such as AI to strengthen democracy (OECD, 2021a).

Methodology

This research used a documentary method, based on a review of the literature to better understand citizen participation, its types, its relationship with artificial intelligence, and how it has helped governments formulate citizen-centered public policies. On that basis, a search was conducted for cases of countries that have used AI as a tool for better citizen participation, in order to analyze the factors, they share and that have supported the correct implementation of AI, and to identify the differences between countries according to their contexts. To analyze countries that apply AI in their governments, reports from international organizations were reviewed, including the Government AI Readiness Index, the OECD Digital Government Index, the World Bank's Global Program on GovTech and Public Sector Innovation, Global Trends in Government Innovation, scientific articles, and official government sites.

Sample

To delimit the countries analyzed, the results of the Government AI Readiness Index 2024 were used, since they indicate which countries are most prepared for AI implementation in government. The ten highest-scoring countries are, in order, the United States, Singapore, Korea, France, the United Kingdom, Canada, the Netherlands, Germany, Finland, and Australia. As the index also groups countries by region, the three top-ranked regions were selected: North America (United States and Canada), Western Europe (France and United Kingdom), and East Asia (Singapore and Korea).

Results

The following analysis examines the leading countries that have used AI in their governments and how they have implemented it, including cases that show how AI has helped them improve citizen participation, together with the contrasting panorama of Latin America.

Leading countries in North America

North America ranks first in government AI readiness. The 2024 index indicates that the United States is the leader, with solid foundations in governance and ethics, innovation capacity, and a mature technology sector that position it as an advanced AI ecosystem. Canada, though not yet at the same technological level, stands out for effective governance and ethics and for data availability, reflecting a focus on responsible AI governance and robust frameworks (Oxford Insights, 2024).

United States

The United States is the leading country in technology and among the most advanced in AI, which it uses to drive industry, empower workers, and strengthen national security through investment in programs that yield high-impact results; it aims to remain the global leader given AI's significant effect on its economy and security (White House, 2025). Its trajectory includes the 2017 G7 outcome document on AI, the 2018 Statement on Artificial Intelligence, and, in 2019, joining other OECD countries in the Recommendation on AI, whose principles the G20 adopted. In 2020 it signed the United States–United Kingdom Declaration on Cooperation in AI R&D and helped launch the Global Partnership on AI (White House, 2025).

By 2021 the White House had established the Office of the National Artificial Intelligence Initiative under the National AI Initiative Act of 2020, creating a focal point for federal coordination on AI research and policy. In October 2023, President Biden issued an executive order to regulate AI in a safe and reliable manner, and many state governments followed with similar regulations (McKinsey & Company, 2024). In 2024 the first National Security Memorandum on AI was issued (Oxford Insights, 2024), and the White House Office of Management and Budget advanced the first government-wide policy to leverage AI and mitigate risks (Downie & O'Brien, 2024). Taking Biden's order as a reference, governors in California, Oklahoma, and Virginia issued guidelines for generative AI addressing operational, informatics, labor, and risk areas (McKinsey & Company, 2024).

A notable application is the launch of ChatGPT Gov, a government-tailored version that lets U.S. agencies use the latest OpenAI models while managing information in a commercial Microsoft Azure cloud, easing data handling with greater security, privacy, and regulatory compliance (OpenAI, 2025). Since its 2024 implementation, more than 3,500 federal, state, and local agencies have sent 18 million messages. For example, the State of Minnesota's Office of Business Translations used AI to deliver faster, more accurate translations at lower cost, while the Commonwealth of Pennsylvania piloted ChatGPT Enterprise to reduce the time required for routine tasks such as analyzing project requirements (OpenAI, 2025). The United States is also applying AI in transportation, healthcare, manufacturing, and financial services (White House, 2025), and conducts digital voter registration through the Vote.gov portal (Vote.gov, 2025).

Canada

Canada, while not matching U.S. leadership, stands out globally and has worked to strengthen its AI governance framework. Through its official portal, it offers several consultation initiatives: the "Connect with government" and "Consulting with Canadians" options let citizens take part in government consultations and express their views, with 860 open consultations reported (Government of Canada, 2024). One example is the public consultation held from November 2023 to January 2024 on Canada's participation in Council of Europe treaty negotiations on AI, which sought citizens' opinions to help develop an inclusive negotiation strategy (Government of Canada, 2025).

Canada also uses chatbots to guide citizens through government services—such as the Passport Chatbot, tested for a limited period to answer questions about the national passport service, and the My Service Canada (MSCA) chatbot, which helps citizens register an account to access government services and benefits; both were trials for which the government disclaimed responsibility, and any personal information entered was deleted (Government of Canada, 2021; Government of Canada, 2024). AI is used mainly for the analysis of opinions: in 2019 the government created the Directive on Automated Decision-Making to improve service delivery, ensuring that automated systems reduce risks, make decisions more efficient and compliant with Canadian law, assess the impact of the algorithms used, and make the resulting data accessible to the public (Government of Canada, 2023).

Canada has worked on AI in government since 2016, with milestones including a technical report (2016–2017), an AI Day event in 2018, a judicial-AI working group, the 2019 Directive on Automated Decision-Making, 2023 guidance on generative AI, the creation of the Canadian Artificial Intelligence Safety Institute (2024) and an Artificial Intelligence Advisory Council (2025), and the publication of the AI Strategy for the Federal Public Service 2025–2027 (Government of Canada, 2025). These efforts reflect a commitment to automating services and decision-making in line with the principles of administrative law: transparency, accountability, legality, and procedural fairness.

Leading countries in Western Europe

Western Europe ranks second in government AI readiness, with several countries—the Netherlands, Germany, and Finland—in the top ten, though France and the United Kingdom give the region its strongest presence. The region stands out for effective data management and infrastructure, consolidated governance, and a solid foundation for AI innovation (Oxford Insights, 2024).

France

In France, interest in smart urban governance has grown in administrative, organizational, and political circles, driven by sustainability demands and the need for greater citizen participation in urban policy. Gourgues et al. (2021) note that participation is inherent to a process of professionalization in France, through the creation of administrative services dedicated to managing participation. France belongs to the Weimar Triangle—alongside Germany and Poland—through which the three countries seek a political alliance to develop national AI investment plans and expand collaboration with the European community (Oxford Insights, 2024).

France has used participation to improve public services, involving civil society in their development (OECD, 2024). For instance, the Ministry of Finance organizes hackathons on tax and policy issues so that participating citizens help improve the government's algorithms and applications; with the information obtained, developers create new services—a solution that has worked well enough to be extended to other ministries to process the large volumes of information received through online platforms (OECD, 2023a).

United Kingdom

The United Kingdom is advancing AI development, having invested £300 million in a computing cluster for research and development and, in 2025, developed the AI Opportunity Action Plan to boost adoption (Oxford Insights, 2024). The UK has used AI to involve citizens in government decisions, as in the European Union panel on virtual worlds, where randomly selected citizens from the 27 member states proposed measures to create attractive and fair European virtual worlds (OECD, 2025a). Some European municipalities have used AI to optimize online citizen input: Cambridge, in the UK, uses the GoVocal platform—developed by a civic-technology startup—for resident consultation, helping officials process thousands of contributions and classify them by location, demographic traits, or emerging issues; its use cut the time required to process citizen requests by 50% (OECD, 2025a).

Leading countries in East Asia

In 2024, East Asia ranked third as a region, with Singapore and Korea among the top performers. These countries stand out for strong governance frameworks, a strategic vision for AI, and solid data availability and infrastructure, though they need more investment in innovation and AI maturity. China, despite its global prominence, does not appear among the top places because of limited data availability and reduced accessibility of public information (Oxford Insights, 2024).

Singapore

Singapore, a highly developed city-state, is distinguished by a national AI strategy that prioritizes scalable, high-impact solutions to increase productivity and encourage citizen participation. The nationwide Pair suite platform, designed to automate administrative tasks, reflects its leadership in improving employee experience and modular government management, while the customization of AI models to local needs shows an approach that balances personal-data protection with efficient execution (Aldemir & Uçma Uysal, 2025). As it consolidated economic stability and social cohesion, Singapore adopted digital mechanisms to foster participation through platforms such as the National Conversation initiative, the Reaching Everyone for Active Citizenry @ Home (REACH) portal, and the Smart Nation program (Soon & Soh, 2014; Shamsuzzoha et al., 2021; Ataman et al., 2025).

The Moments of Life initiative, integrated within Smart Nation, leverages AI and digital technologies to simplify access to public services at key life stages—such as the birth of a child—allowing parents to register a birth, manage financial aid, and schedule vaccinations, strengthening the link between citizens and state institutions (Huiling & Goh, 2017; Aldemir & Uçma Uysal, 2025). Launched in 2014 and continually expanding, Smart Nation integrates technological and data-driven solutions to enhance competitiveness, sustainability, and quality of life, supported by significant investment in digital infrastructure such as the Government Commercial Cloud (Hartley & Aldag, 2024). According to Lee (2024), the pilot AI-Based Citizen Question-Answer Recommender System (ACQAR) advanced the quality of citizen services by raising satisfaction and helping staff respond more promptly and accurately to inquiries.

Other initiatives where AI is already applied include Pulse of the Economy, which merges data sources to monitor economic activity and mobility in real time and identify growth opportunities; Ask Jamie, a virtual assistant integrated into agency portals that uses natural-language processing to answer questions across agencies and can scale to live chat; and KeyReply, the first government chatbot integrated into Facebook Messenger, which routes citizens' queries to the appropriate department (Susar & Aquaro, 2019). The Analytics and Artificial Intelligence Lab, affiliated with the Defence Science and Technology Agency, develops tools using real-time data from the Internet of Things, cameras, and sensors to detect security threats. In this way, Singapore positions itself as a global benchmark in advanced urban management.

South Korea

South Korea is recognized as a global leader in information technology and digital government, having achieved one of the most significant advances in digital government over the past five decades (Chung et al., 2022). It has one of the fastest Internet speeds in the world and was the first OECD member to reach 100% wireless broadband coverage, with a growing impact on political dynamics and participation (Hu & Odei, 2023). According to Member States' responses to the United Nations E-Government Survey (2018), more than 40 countries reported using emerging technologies such as AI, the Internet of Things, and blockchain; in 2017, Korea's National Human Resources Development Institute offered 28 courses to more than 2,000 public servants to strengthen ICT competencies, including content on big data, AI, and the application of emerging technologies in public administration (Susar & Aquaro, 2019).

In 2017, South Korea launched the Gwanghwamun 1st Street platform to foster direct democracy by allowing citizens to propose public-policy ideas, inspiring the Gwanak-gu district's own online office built on active user participation. In 2020 it introduced a specific impact assessment for AI, becoming the first country to establish one in national legislation (Jeong, 2022). To address COVID-19, it implemented a contact-tracing program combining traditional epidemiological methods with GPS data, credit-card transactions, and surveillance records, enabling more efficient detection of contagions (Park et al., 2020). In May 2023 it published guidelines for the use of ChatGPT in government, establishing a framework for incorporating generative AI into public operations across functions such as information retrieval, translation, analysis, text generation, and summarization (Beltran et al., 2024). Korea has also implemented Metaverse Seoul, which lets citizens access administrative services through avatars in a three-dimensional virtual space covering culture, economy, education, and tourism (OECD, 2025a). These initiatives reflect a commitment to the safe integration of emerging technologies, supported by strong digital legislation and efforts to reduce the digital divide.

Countries in Latin America and the Caribbean

Although certain countries lead the world in government AI, others are working on implementation, with Colombia and Uruguay standing out as regional leaders thanks to well-designed strategies already under way (OECD/CAF, 2022). Uruguay is considered a leader for its e-government initiatives, sound data-protection regulation, and good Internet access, while Argentina, Brazil, and Peru show strong commitment and execution but lack the digital maturity of the leading countries. Mexico, the first country in the region to publish a national AI strategy, lost its early leadership for want of clear priorities, while Mexico and Costa Rica still show high commitment. The countries with the least adoption capacity are Barbados and Panama; the Dominican Republic, Ecuador, Jamaica, and Paraguay have advanced indirectly through data-protection laws; and Bolivia, Trinidad and Tobago, and Venezuela lag the furthest behind (OECD/CAF, 2022).

Compared with more developed countries, Latin America faces prior concerns—being one of the most unequal regions in the world, marked by economic and social crises, corruption, impunity, and a weak rule of law—that complicate the adoption of emerging technologies. In these contexts, technology companies have approached governments with a discourse of innovation and progress, promising greater efficiency and security; where regulation is absent or lax, companies that develop AI gain entry with the promise of improving citizens' lives, sometimes at no cost (Ricaurte et al., 2024). The following cases show how AI developed by private companies and implemented in specific contexts has not always succeeded, sometimes deepening the exclusion of vulnerable groups.

Chile

With Microsoft, Chile implemented the social program Alerta Niñez to predict which low-income adolescents had a higher propensity to commit crimes, in order to design supportive public policies. Using such predictive technologies deepened exclusion and discrimination against this group, as it can generate inaccurate predictions, lacks transparency in how it works, uses data without consent, risks leaking personal data, and stigmatizes a vulnerable sector of society (Ricaurte et al., 2024).

Brazil

Brazil worked with Microsoft to develop a system, through the National Employment System (SINE), to relocate unemployed people in the labor market by generating worker profiles and offering personalized job and training opportunities. One problem was that accessing the platform required Internet, which nearly a quarter of Brazil's population lacks (Velasco & Venturini, 2021).

Conclusions

Citizen participation is a key element of government: it helps formulate public policies and enables citizens to regain trust and feel part of the governance developed in their territory. To facilitate it, governments must invest in innovation and infrastructure that bring them closer to the population—something that emerging technologies such as AI can enable by making collaboration between governments and citizens easier.

Despite the much-discussed benefits of AI in government, many challenges remain. The analysis shows that applying AI is a difficult path: governments must first understand how it works and, above all, regulate it. The world's leading countries and regions are still working to understand it and face significant challenges even with adequate infrastructure and budgets, while economically disadvantaged countries lag in this technological change. According to the literature, the application of AI in citizen participation appears to be in its infancy, since governments use it in a one-directional way—mainly to inform or consult—and rarely reach the point where citizens participate in all stages of formulating the policies that benefit them.

Among the countries analyzed, North America and Western Europe have done the most to build legal frameworks responding to technological change; yet their use of AI in participation is largely confined to information and consultation, without deeper integration of society into government decisions. East Asia is no exception, though it uses AI in friendlier ways—as in Seoul's metaverse avatars—while lagging behind in technological infrastructure and AI-related legal frameworks.

Although the United States leads in technology and AI, in the 2021 OECD study its inhabitants did not feel they had a say in government decisions, so it must continue to strengthen participation alongside AI. Canada, despite lacking the same infrastructure, scored above the OECD average in citizens feeling considered in decision-making, as did the United Kingdom, suggesting that these two countries maintain a good relationship with citizen participation. In Latin America and the Caribbean, leaders such as Colombia and Uruguay have taken firm steps and advanced toward data-protection policies; yet AI, while easing some processes, has also widened the gap of inequality and exclusion.

Emerging technologies such as AI have helped governments reduce bureaucratic processes, encourage participation, and follow up on procedures in a more automated way, but they have also highlighted inequalities, since participation platforms often do not consider the context of the people affected by public-policy decisions—leaving decisions for the most vulnerable in the hands of those with better contexts. Although economic, geographic, and educational differences have always existed, AI has made them more visible, as governments assume that these technologies will bring them closer to people without providing the conditions for vulnerable groups to access them.

Therefore, citizen-participation processes should adopt a perspective that offers equal conditions for all participants, focusing not only on results but on making the entire process accessible regardless of context. Rather than addressing every government problem through a technological lens, unresolved social and economic issues should also be tackled. Governments should provide the tools to make these processes easy and natural, using colloquial, understandable language so that participation is well received and valued.

References

Aldemir, C., & Uçma Uysal, T. (2025). Artificial Intelligence for Financial Accountability and Governance in the Public Sector: Strategic Opportunities and Challenges. Administrative Sciences, 15(2), 58.

Ataman, C., Herthogs, P., Tunçer, B., & Perrault, S. (2025). From insight to action: An integrated assessment framework for digital citizen participation in data-centric urban practices. Cities, 156, 105545.

Balduzzi, T., & Siclari, L. (2024). Participative and Deliberative Democracy Facing Technology: A Study on Digital Democratic Innovations. Rivista italiana di informatica e diritto, 6(1), 87-118.

Beltran, M. A., Ruiz Mondragon, M. I., & Han, S. H. (2024, June). Comparative analysis of generative AI risks in the public sector. In Proceedings of the 25th Annual International Conference on Digital Government Research (pp. 610-617).

Bingöl, E. (2022). Citizen Participation in Smart Sustainable Cities. In I. Management Association (Ed.), Research Anthology on Citizen Engagement and Activism for Social Change. pp. 967-987. IGI Global Scientific Publishing. https://doi.org/10.4018/978-1-6684-3706-3.ch052

Bonsón, E., Royo, S., & Ratkai, M. (2015). Citizens' engagement on local governments' Facebook sites. An empirical analysis: The impact of different media and content types in Western Europe. Government Information Quarterly, 32, 52–62.

Brown, D. (2005). Electronic government and public administration, International Review of administrative science. Int. Rev. Adm. Sci. 71.241-254.

Castells, M. (2011). The rise of the network society. Hoboken, NJ: John Wiley.

Chen, T., Guo, W., Gao, X., & Liang, Z. (2021). AI-based self-service technology in public service delivery: User experience and influencing factors. Government Information Quarterly, 38(4), 101-520.

Chung, C. S., Choi, H., & Cho, Y. (2022). Analysis of digital governance transition in South Korea: Focusing on the leadership of the president for government Innovation. Journal of Open Innovation: Technology, Market, and Complexity, 8(1), 2.

Citizen Participation Forum. (2021). Citizen Participation Index: Methodology. Link: https://index.fgu.bg/en/methodology.html

Coeckelbergh, M. (2022). Democracy, epistemic agency, and AI: Political epistemology in times of artificial intelligence. AI and Ethics, 1–10. https://doi.org/10.1007/s43681-022-00239-4

Contreras, P., & Montecinos, E. (2019). Democracia y participación ciudadana: Tipología y mecanismos para la implementación. Revista de Ciencias Sociales. 25 (2). 178-191.

Criado, J. I. (2016). Las administraciones públicas en la era del gobierno abierto. Gobernanza inteligente para un cambio de paradigma en la gestión pública. Revista de Estudios Políticos. 173, 245-275. https://doi.org/10.18042/cepc/rep.173.07

Danaher, J. (2016). The threat of algocracy: Reality, resistance and accommodation. Philosophy & Technology, 29(3), 245–268. DOI: https://doi.org/10.1007/s13347-015-0211-1

Downie, S., & O'Brien. K. (December 20, 2024). IA en el gobierno. IBM Link:

https://www.ibm.com/mx-es/think/topics/ai-in-government

Elstub, S. & Escobar, O. (2019). Handbook of democratic innovation and governance. Edward Elgar Publishing

Engstrom, D. F., Ho, D. E., Sharkey, C. M., & Cuéllar, M. F. (2020). Government by algorithm: Artificial intelligence in federal administrative agencies. NYU School of Law, Public Law Research Paper, (20-54).

European Commission. (2024). Adopt AI Study. Final study report. Directorate General for Communications Networks, Content and Technology (CNECT). European Commission. https://doi.org/10.2759/22251

Fernández, J. V. (2023). Artificial intelligence in government: Risks and challenges of algorithmic governance in the administrative state. Ind. J. Global Legal Stud., 30, 65.

Fischer F (1990) Technocracy and the Politics of Expertise. Newbury Park, CA: Sage Publishing.

Gesk, T. S., & Leyer, M. (2022). Artificial intelligence in public services: When and why citizens accept its usage. Government Information Quarterly. 39(3), 101-704.

Giffinger, R., Fertner, C., Kramar, H., Kalasek, R., & Pichler, N. (2007). Smart cities: Ranking of European medium-sized cities. Vienna University of Technology.

Gilman, H. R. & Peixoto, T. C. (2019). Digital participation. In S. Elstub & O. Escobar (Eds.), Handbook of Democratic Innovation and Governance.Edward Elgar Publishing

Gourgues, G., Mazeaud, A., & Norjon, M. (2021). From the participatory turn of administrations to the bureaucratisation of participatory democracy: study based on the French case. International Review of Administrative Sciences. 88(4). 1141-1158. https://doi.org/10.1177/00208523211003

Government of Canada. (April 25, 2023). Directive on Automated Decision-Making Government of Canada. Link: https://www.tbs-sct.canada.ca/pol/doc-eng.aspx?id=32592

Government of Canada. (April 25, 2024). Terms of use for MSCA chatbot. Government of Canada. Link: https://www.canada.ca/en/employment-social-development/services/my-account/terms-use-chatbot.html

Government of Canada. (April 3, 2025). Share your views: Canada's participation in treaty negotiations on artificial intelligence at the Council of Europe. Government of Canada.

Government of Canada. (July 31, 2024). Connect with government. Government of Canada. Link: https://www.canada.ca/en/government/system/consultations.html

Government of Canada. (March 04, 2025). Progress on AI in government. Government of Canada. Link: https://www.canada.ca/en/government/system/digital-government/digital-government-innovations/responsible-use-ai/progress.html

Government of Canada. (March 07, 2025). Advisory Council on Artificial Intelligence. Government of Canada. Link: https://ised-isde.canada.ca/site/advisory-council-artificial-intelligence/en

Government of Canada. (May 22, 2021). Terms of Use for Passport Chatbot. Government of Canada. Link: https://www.tbs-sct.canada.ca/pol/doc-eng.aspx?id=32592

Hartley, K. (2023). Public perceptions about smart cities: governance and quality-of-life in

Hong Kong. Soc Indic Res. 166. 731–753.

Hartley, K., & Aldag, A. (2024). Public trust and support for government technology: Survey insights about Singapore's smart city policies. Cities, 154, 105-368.

Hovik, S., Legard, S., McShane, I., Middha, B., Reichborn-Kjennerud, K., & Ruano, J. M. (2022). Participation and Influence in Urban Development: Does City E-Participation Strategy Matter? In Citizen Participation in the Information Society (No. 2, pp. 25-47). Palgrave Macmillan.

Hu, H., & Odei, S. A. (2023). Enlightenment of foreign citizens' online political participation on China—Based on the case study of America, Singapore, South Korea, and Japan. Development, 7(3), 26-80.

Huiling, E., & Goh, B. (2017). AI, robotics and mobility as a service: The case of Singapore. Field Actions Science Reports. The Journal of Field Actions, (Special Issue, 17), 26-29.

International Association for Public Participation (IAP2). (2024). IAP2 Spectrum of Public Participation. IAP2. https://www.iap2.org/page/SpectrumEvolution

Jäntti, A., & Kurkela, K. (2021). How municipalities can enhance citizen participation? – Exploring the views of participants and non-participants. Scandinavian Journal of Public Administration, 25(1), 23-42.

Jäntti, A., Paananen, H., Kork, A. A., & Kurkela, K. (2023). Towards interactive governance: embedding citizen participation in local government. Administration & society. 55(8).1529-1554.

Jeong, J. (2022). Introduction of the first AI impact assessment and future tasks: South Korea discussion. Laws. 11(5), 73.

Kouziokas, G. N. (2017). The application of artificial intelligence in public administration for forecasting high crime risk transportation areas in urban environment. Transportation research procedia, 24, 467-473.

Lahdili, N., & Nyadera, I. N. (2024). Artificial Intelligence and Citizen Participation in Governance: Opportunities and Threats. Amme Idaresi Dergisi, 57(3).

Lan, L., Zhu, Q., & Liu, Y. (2021). The rule of virtue: a Confucian response to the ethical challenges of technocracy. Science and Engineering Ethics, 27(5), 64. The rule of virtue: a Confucian response to the ethical challenges of

Landemore, H. (2022). Can AI bring deliberative democracy to the masses. In Human Centered Artificial Intelligence Seminar, Stanford University.Can AI bring deliberation to the masses. Retrieved from https://www.law.nyu.edu/sites/default/files/Helen%20Landemore%20Can%20AI%20bring%20deliberative%20democracy%20to%20the%20masses.pdf

Lee, H. S. (2024). Implementation and evaluation of AI-based Citizen Question-Answer Recommender (ACQAR) to enhance citizen service delivery in Singapore public sector: A case study.

Li, Y., Fan, Y., & Nie, L. (2023). Making governance agile: Exploring the role of artificial intelligence in China's local governance. Public Policy and Administration, 09520767231188229.

Lindgren, I., Janssen, M., Lee, H., Polini, A., Rodríguez, M., Jochen, H. & Tambouris, E. (2019).

Electronic Government. 18th IFIP WG 8.5 International Conference, EGOV San Benedetto Del Tronto, Italy. Sprinter. http://doi.org/10.1007/978-3-030-27325-5.

Lykidis, I., Drosatos, G. & Rantos, K. (2021). The Use of Blockchain Technology in e-Government Services. Computer. 10. 168.

Madan, R., & Ashok, M. (2022). A public values perspective on the application of Artificial Intelligence in government practices: A Synthesis of case studies. In Handbook of research on artificial intelligence in government practices and processes. IGI Global Scientific Publishing.

Mariñez, F. (2022). Participación ciudadana colaborativa y los valores de la administración pública. Revista de Gestión Pública. 11(2), 185–203. https://doi.org/10.22370/rgp.2022.11.2.3715

McKinsey and Company. (21 de febrero de 2024). El despliegue de IA generativa en los gobiernos estatales de EE.UU.: Poner a prueba, escalar y adoptar. McKinsey and Company. Link: https://www.mckinsey.com/featured-insights/destacados/el-despliegue-de-

Meijer A, Lorenz L, & Wessels M (2021) Algorithmization of bureaucratic organizations: using a practice lens to study how context shapes predictive policing systems. Public Administration Review. 81(5): 837–846.

Mellouli, S., Janssen, M., & Ojo, A. (2024). Introduction to the Issue on Artificial Intelligence in the Public Sector: Risks and Benefits of AI for Governments. Digital Government: Research and Practice, 5(1), 1-6. https://doi.org/10.1145/3636550

Mohammed, A. K. (2013). Civic Engagement in Public Policy Making: Fad or Reality in G hana?. Politics & Policy, 41(1), 117-152.

Nabatchi, T. & Amsler, L. B. (2014). Direct public engagement in local government. The American Review of Public Administration, 44(4), 63-88.

OECD. (2017). Recomendación del Consejo sobre Gobierno Abierto. OECD. Link: https://legalinstruments.oecd.org/en/instruments/OECD-LEGAL-0438

OECD. (2021a). Innovative public participation. OECD. https://www.oecd.org/en/topics/sub-issues/open-government-and-citizen-participation/innovative-public-participation.html.

OECD. (2021b). Open government and citizen participation. OECD. https://www.oecd.org/en/topics/open-government-and-citizen-participation.html

OECD. (2022). Directrices de la OCDE sobre Procesos de Participación Ciudadana. Recuperado deOECD. Link: https://www.oecd.org/content/dam/oecd/es/publications/reports/2022/09/oecd-guidelines-for-citizen-participation-processes_63b34541/f1b22902-es.pdf

OECD (2023a), Directrices de la OCDE sobre Procesos de Participación Ciudadana, Estudios de la OCDE sobre Gobernanza Pública, OECD Publishing, Paris, https://doi.org/10.1787/f1b22902-es.

OECD. (2023b). 2023 OECD Digital Government Index. OECD. Link: https://www.oecd.org/en/publications/2023-oecd-digital-government-index_1a89ed5e-en.html

OECD. (2024). Global Trends in Government Innovation 2024. OECD. Link: https://www.oecd.org/en/publications/global-trends-in-government-innovation-2024_c1bc19c3-en.html

OECD: (2025a), Tackling civic participation challenges with emerging technologies: Beyond the hype. Paris: OECD. https://doi.org/10.1787/ec2ca9a2-en

OECD. (2025b). Artificial intelligence. https://www.oecd.org/en/topics/artificial-intelligence.html

OECD/CAF. (2022). Uso estratégico y responsable de la inteligencia artificial en el sector público de América Latina y el Caribe, Estudios de la OCDE sobre Gobernanza Pública, OECD. https://doi.org/10.1787/5b189cb4-es (https://doi.org/10.1787/5b189cb4-es).

OpenAI. (January 28, 2025). Introducing ChatGPT Gov. OpenAI. Link: https://openai.com/global-affairs/introducing-chatgpt-gov/

Ospina, D., M, R. & Zambrano, O., K, J. (2023). Gobierno digital e inteligencia artificial, una mirada al caso colombiano. Administración & Desarrollo, 53(1), 1-34. https://doi.org/10.22431/25005227

Oxford Insights. (2024). Government AI Readiness Index 2024. Oxford Insights. Link: https://oxfordinsights.com/ai-readiness/ai-readiness-index/

Palak Dev, & Sainger, G. (2025). Benefits of Digital Training Approaches on Non-Banking Economic Divisions in India. Mercados y Negocios, (55), 3–32. https://doi.org/10.32870/myn.vi55.7771

Park, Y. J., Choe, Y. J., Park, O., Park, S. Y., Kim, Y. M., Kim, J., & Jeong, E. K. (2020). Contact tracing during coronavirus disease outbreak, South Korea, 2020. Emerging infectious diseases, 26(10), 2465.

Pislaru, M., Vlad, C. S., Ivascu, L., & Mircea, I. I. (2024). Citizen-Centric Governance: Enhancing Citizen Engagement through Artificial Intelligence Tools. Sustainability, 16(7), 2686.

Puerta, M., Pindado, J., & González, E. (2023). The resilience of corporate tourism: bleisure, digitalization, and sustainability. Mercados y Negocios, (48), 53–74. https://doi.org/10.32870/myn.vi48.7679

Ricaurte, P., Gómez-Cruz, E., & Siles, I. (2024). Gubernamentalidad algorítmica en América Latina: Imaginarios sociotécnicos, poder blando neocolonial y autoritarismo. Big Data & Society, 11 (1). https://doi.org/10.1177/20539517241229697

Romberg, J., & Escher, T. (2024). Making Sense of Citizens' Input through Artificial Intelligence: A Review of Methods for Computational Text Analysis to Support the Evaluation of Contributions in Public Participation. Digital Government: Research and Practice, 5(1), 1-30. https://doi.org/10.1145/3603254

Ruano, J. M., & Reichborn-Kjennerud, K. (2022). Inside the black box: Perspectives and attitudes of civil servants on citizen participation. In Citizen participation in the information society: Comparing participatory channels in urban development. Cham: Springer International Publishing.

Shamsuzzoha, A., Nieminen, J., Piya, S., & Rutledge, K. (2021). Smart city for sustainable environment: A comparison of participatory strategies from Helsinki, Singapore and London. Cities, 114, 103194.

Sharma, & Padashetty, S. (2025). Impact of E-Service Quality of Mobile Application on Customer Perceived Value. Mercados y Negocios, (55), 33–62. https://doi.org/10.32870/myn.vi55.7818

Soon, C., & Soh, Y. D. (2014). Engagement@ web 2.0 between the government and citizens in Singapore: dialogic communication on Facebook? Asian Journal of Communication, 24(1), 42-59.

Steinbach, M., Sieweke, J., & Süß, S. (2019). The diffusion of e-participation in public administrations: A systematic literature review. Journal of Organizational Computing and Electronic Commerce, 29(2), 61–95.

Susar, D., & Aquaro, V. (2019). Artificial intelligence: Opportunities and challenges for the public sector. In Proceedings of the 12th International Conference on Theory and Practice of Electronic Governance (pp. 418-426).

Tangi, L., Noordt, C. van., Combetto Marco., Gattwinkel, D., & Pignatelli, F. (2022). AI watch. European Landscape on the Use of Artificial Intelligence by the Public Sector. Luxembourg: Publications Office of the European Union.

United Nations Department of Economic and Social Affairs. (2018). United Nations. E-Government Survey 2018. Uescap. Link: https://www.unescap.org/sites/default/files/E-Government%20Survey%202018_FINAL.pdf

Van Noordt, C., & Misuraca, G. (2022). Artificial intelligence for the public sector: results of landscaping the use of AI in government across the European Union. Government information quarterly, 39(3), 101714.

Vote.gov. (2025). Register for voting. U.S. General Services Administration. Link: https://vote.gov/es (https://vote.gov/es)

Vrabie, C. (2023). E-Government 3.0: An AI Model to Use for Enhanced Local Democracies. Sustainability, 15(12), 9572. https://doi.org/10.3390/su15129572 (https://doi.org/10.3390/su15129572)

WEF. (2024). Generative AI Governance: Shaping a Collective Global Future. WEF. Link: https://www3.weforum.org/docs/WEF_Generative_AI_Governance_2024.pdf

White House. (2025). Artificial for the American People. White House. Link: https://trumpwhitehouse.archives.gov/ai/ai-american-innovation/

Whitford, A. B., Yates, J., Burchfield, A., Anastasopoulos, J. L., & Anderson, D. M. (2020). The adoption of robotics by government agencies: evidence from crime labs. Public Administration Review, 80(6), 976-988.

World Bank. (2022). The GovTech Maturity Index (GTMI). World Bank. Link: https://openknowledge.worldbank.org/server/api/core/bitstreams/5e157ee3-e97a-5e42-bfc0-f1416f3de4de/content

Zheng, Y., Schachter, H. & Holzer, M. (2014). The impact of government form on e-participation: A study of New Jersey municipalities. Government Information Quarterly, 31. 653-659.

Zhu, Y., Janssen, M., Wang, R., & Liu, Y. (2022). It is me, chatbot: working to address the COVID-19 outbreak-related mental health issues in China. User experience, satisfaction, and influencing factors. International Journal of Human–Computer Interaction, 38(12), 1182-1194.

Zidouemba, M. T. (2025). Governance and artificial intelligence: the use of artificial intelligence in democracy and its impacts on the rights to participation. Discover Artificial Intelligence, 5(1), 1-11.

Zuiderwijk, A., Chen, Y. C., & Salem, F. (2021). Implications of the use of artificial intelligence in public governance: A systematic literature review and a research agenda. Government information quarterly, 38(3), 101-577.